

NAGA SATYA PHANINDRA VALLABHANENI

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PROFESSIONAL SUMMARY

Analytical Python Engineer with experience in software testing, automated workflow development, data analysis, system telemetry, debugging, and performance evaluation. Experienced in developing Python-based automation and data pipelines, analyzing complex datasets and system metrics, creating data visualizations and performance reports, and performing root-cause analysis across distributed software environments. Research experience includes time-series environmental data, machine learning, edge-case analysis, and performance evaluation. Strong analytical, detail-oriented, and collaborative problem-solving skills.

CORE COMPETENCIES & TECHNICAL SKILLS

- **Programming & Automation:** Python, FastMCP, REST APIs, SQL, Workflow Automation, Data Pipeline Design
- **Testing, Triage & Debugging:** Software Testing, Integration Testing, Regression Testing, Root-Cause Analysis, Software Triage, Debugging, Edge-Case Analysis, Fault-Tolerance Analysis
- **Data Analysis & Performance Evaluation:** Time-Series Analysis, Sensor Data Analysis, System Telemetry, Performance Metrics, Coverage Analysis, Data Validation, Statistical Analysis, Performance Reporting
- **Data Visualization & Machine Learning:** Matplotlib, Scikit-learn, TensorFlow, PyTorch, LSTM, CNN, Random Forest, BERT, Feature Engineering
- **Systems & Infrastructure:** Docker, Kubernetes, Jenkins, GitHub Actions, Git, CI/CD, AWS, GCP
- **Collaboration:** Technical Documentation, Cross-Functional Collaboration, Agile/Scrum, Code Reviews, Problem Solving

TECHNICAL EXPERIENCE

University of Dayton, Dayton, OH — AI Solutions Associate Dec 2025 – May 2026

- Built Python-based data ingestion, normalization, and validation pipelines to process and curate 4,400+ complex institutional data records for automated query routing and reliable system behavior.
- Implemented structured Model Context Protocol (MCP) tool schemas and protocol-based routing to support automated validation, connection logging, real-time execution, and systematic issue analysis.
- Evaluated system performance metrics, test coverage, and response behavior, generating analytical performance reports that improved response accuracy to 88% and reduced loss by 50.6%.
- Analyzed edge cases and system behavior across complex data conditions to identify inconsistencies, validate expected outputs, and improve workflow reliability.

Accenture, Hyderabad, India — Infrastructure & Python Engineer Oct 2022 – Jul 2024

- Performed software triage, debugging, fault-tolerance analysis, and root-cause investigation on complex data integration workflows, identifying edge cases and resolving issues affecting system performance.
- Developed Python automation scripts and high-throughput data pipelines processing 3.5M+ daily records, automating data workflows and metadata extraction while maintaining high data fidelity.
- Designed and executed comprehensive QA, integration, and regression testing suites across distributed services, reducing production software bugs by 80%.
- Validated inter-service communication, payload schemas, system telemetry, and expected system behavior across 14 enterprise environments, maintaining 99.9% uptime.
- Streamlined release and testing workflows using Jenkins, Docker, and GitHub Actions, transitioning deployment workflows from weekly to daily releases.
- Collaborated with engineering teams through code reviews, pair programming, technical troubleshooting, and mentoring junior developers.

LJ Technologies, Hyderabad, India — Systems & Infrastructure Engineer May 2021 – Jul 2022

- Developed scalable Python automation pipelines for data extraction, schema validation, and metadata normalization across backend systems.
- Conducted comprehensive unit, integration, and regression testing across distributed API services to ensure system stability and data consistency.

- Maintained CI/CD deployment pipelines using Docker, Git, and Jenkins to streamline software release workflows and maintain environment stability.

TECHNICAL PROJECTS

Real-Time Streaming Feature Store — GitHub · Python, FastAPI, Docker Sep 2026

- Built a real-time feature-serving service in Python/FastAPI computing tumbling-window count/sum/avg aggregations over event streams, with watermark-based late-event handling and TTL state eviction.
- Implemented a pluggable online store (in-memory and SQLite backends), backfill from unsorted historical events, a seeded synthetic event generator, and a serverless CLI demo.
- Verified with 59 pytest tests, ruff-clean code, Docker Compose deployment, and a passing GitHub Actions CI pipeline.

Real-Time Log Anomaly Detection Service — GitHub · Python, FastAPI, Scikit-learn, Docker Sep 2026

- Built a log anomaly detection service using IsolationForest over 16 engineered features (log-template mining, latency statistics), exposed via /train, /score, and /ingest REST endpoints.
- Achieved 86.8% recall on 38 injected anomalies in seeded synthetic data; shipped with a synthetic log generator and CLI demo.
- Verified with 19 pytest tests, Docker packaging, and a passing GitHub Actions CI pipeline.

Plant Yield and Growth Prediction Using ML and DL Algorithms — IEEE ICEARS 2022 2022

- Developed Python analysis workflows using LSTM, CNN, Random Forest, and SVR models to analyze time-series environmental and growth metrics in controlled environments.
- Evaluated predictive model performance across varying environmental conditions and generated data visualizations and analytical reports to assess model accuracy and sensor data reliability.

Phishing URL Detection Using BERT and Cybersecurity Analysis — Python, BERT, NLP 2022

- Feature-engineered a 12M+ row dataset using Python to build a BERT-based NLP classification pipeline achieving 78% accuracy; open-sourced on GitHub.
- Evaluated model coverage and performance metrics, developing diagnostic scripts to analyze misclassifications and triage edge-case inputs across large datasets.

FlyerGPT AI Platform & MCP Agent Infrastructure — Python, FastMCP, Azure OpenAI 2025

- Architected an AI assistant platform and Model Context Protocol (MCP) agent infrastructure using Python, FastMCP, and Azure OpenAI for conversational database access.
- Implemented structured tool schemas, access controls, and connection event logging to validate data queries and ensure reliable system behavior.

EDUCATION

- **Master of Science in Computer Science** — University of Dayton, May 2026
- **Bachelor of Technology in Electrical & Communications Engineering** — GRIET, May 2022

CERTIFICATIONS

- GitHub Foundations Badge (2025) · NPTEL: Joy of Computing Using Python · Oracle AI Foundations Associate